ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 20 m/s	(150+250)/20.
2	**(b)** 15 km	60 - 45.
3	**(b)** 1.82 s	Mean 18.2 / 10.
4	**(d)** 300 m	300/30.
5	**(d)** 360 km	Gap 120 km, closing 30 km/h in 4 h.
6	**(d)** less than true by about 0.2 s	Late start.
7	**(d)** 4 m/s	400/100.
8	**(c)** 12 km	x/4 + x/6 = 5 gives x = 12.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the relation between minute and second does not explain why the second is the SI unit.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R restates the effect of length on the time period and explains A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distance 500 m in 30 s (1); 500/30 = 16.67 m/s, i.e. 60 km/h (1).	1+1
12	Time spent at each speed differs (1); Average speed = total distance/total time (1).	1+1
13	The speed stays constant (1); Equal distances are covered in equal times (1).	1+1
14	Mean total = 15.2 s (1); T = 1.52 s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Speed = 120/12 = 10 m/s (1); Distance in 30 s = 300 m (1); Platform = 300 - 120 = 180 m (1).	1+1+1
16	Time = 3 h (1); Distance = speed x time (1); 40 x 3 = 120 km (1).	1+1+1
17	1700/340 (1); 5 s (1); Distance/speed (1).	1+1+1
	OR: Take 120 km each way (1); Times 3 h and 2 h (1); 240/5 = 48 km/h (1).	1+1+1
18	Time period 2 s (1); Each swing 1 s (1); Counting swings (1).	1+1+1
	OR: 25 m/s (1); 25 m (1); Convert (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $60/30 = 2$ s (1). (ii) No, it does not (1). (iii) $50/20 = 2.5$ s (1). (iv) The second pendulum is longer (1), because a longer pendulum has a larger time period (2.5 s is greater than 2 s) (1).	1+1+1+2
20	Lengths 25, 50, 75, 100 cm (1); 20 oscillations timed each (1); T = t/20 (1); T increases with length (1); Mass kept same (1).	1+1+1+1+1
	OR: Distance 5 x 400 = 2000 m (1); Time 10 min = 600 s (1); Speed 2000/600 = 3.33 m/s (1); 3.33 x 18/5 = 12 km/h (1); Speed = distance / time (1).	1+1+1+1+1

Mistakes to avoid:

- Dividing time by the number of oscillations the wrong way; time period = **total time ÷ number of oscillations**.
- Writing that a heavier bob has a larger time period; the period depends on **length**, not on mass.
- Forgetting to convert units before calculating (minutes to hours, km/h to m/s).
- Using $\times \frac{18}{5}$ instead of $\times \frac{5}{18}$ when converting km/h to m/s.
- Calculating average speed as the average of two speeds instead of *total distance divided by total time*.
- Reading a horizontal line on a distance-time graph as 'moving fast'; it means the object is **at rest**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 40 km	8 km/h x 5.
2	**(a)** 0.25 per cent	0.1/40.
3	**(b)** 30 km/h	2x/(x/20 + x/60) = 30.
4	**(a)** the speed is constant but the direction changes	Uniform circular motion.
5	**(d)** the smallest time it can show is 0.01 s	Resolution.
6	**(d)** 2 : 5	10 : 25.
7	**(d)** 12 m	2 x 6.
8	**(b)** equal-time swings	Isochronism for small angles.
9	**(c)** A is true but R is false.	$90/2 = 45$ km/h so A is true; speed = distance / time, so R is false.
10	**(d)** A is false but R is true.	$36 \times \frac{5}{18} = 10$ m/s, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	20 min = 1/3 h (1); 30 x 1/3 = 10 km (1).	1+1
12	T = 45/15 = 3 s (1); 180/3 = 60 oscillations (1).	1+1
13	A runs 8 m/s for 10 s = 80 m (1); 20 m behind (1).	1+1
14	24 x 60 x 60 (1); 86400 s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Car 15 m/s (1); 15 x 300 = 4500 m; bike 3600 m (1); Car ahead by 900 m (1).	1+1+1
16	15 km in 50 min (1); 18 km/h (1); Both parts 18 km/h: uniform (1).	1+1+1
17	Stops (1); Slows at turns (1); Speeds up (1).	1+1+1
	OR: 12 m (1); 0 m then 12 m (1); Total 24 m in 8 s: 3 m/s (1).	1+1+1
18	Reaction time equal (1); Systematic error cancels (1); Better comparisons (1).	1+1+1
	OR: 3 km in 45 min (1); 3/0.75 (1); 4 km/h (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $144/2 = 72$ km/h (1). (ii) $72 \times \frac{5}{18} = 20$ m/s (1). (iii) $72 \times 3.5 = 252$ km (1). (iv) $360/72 = 5$ hours (1); the train is assumed to move with the same (uniform) speed throughout (1).	1+1+1+2
20	2 h and 2 h = 4 h (1); 50 km total (1); 12.5 km/h (1); Simple average 12.5 (1); Equal here because time equal (1).	1+1+1+1+1
	OR: 90 min = 5400 s (1); 1440/90 = 16 orbits (1); T = 5400 s (1); Periodic motion (1); Orbits per day 16 (1).	1+1+1+1+1

Mistakes to avoid:

- Dividing time by the number of oscillations the wrong way; time period = **total time ÷ number of oscillations**.
- Writing that a heavier bob has a larger time period; the period depends on **length**, not on mass.
- Forgetting to convert units before calculating (minutes to hours, km/h to m/s).
- Using $\times \frac{18}{5}$ instead of $\times \frac{5}{18}$ when converting km/h to m/s.
- Calculating average speed as the average of two speeds instead of *total distance divided by total time*.
- Reading a horizontal line on a distance-time graph as 'moving fast'; it means the object is **at rest**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** to-and-fro periodic	Vibrates.
2	**(d)** lengthen the pendulum slightly	A longer pendulum has a larger time period, so the clock slows.
3	**(a)** not possible for distance moved	Distance does not decrease.
4	**(c)** 27 days	Orbital period.
5	**(b)** non-uniform	Speeds differ.
6	**(a)** 85 s	60 + 25.
7	**(b)** 24 h	A day.
8	**(c)** speed	Speed is non-negative.
9	**(a)** Both A and R are true and R is the correct explanation of A.	A straight line means the distance grows equally in equal times, which is uniform speed.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but they are different instruments and neither explains the other.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	T = 40/25 (1); 1.6 s (1).	1+1
12	28 km (1); 28/56 = 0.5 h (1).	1+1
13	340 x 18/5 (1); 1224 km/h (1).	1+1
14	Mark equal intervals of time and measure distances (1); Equal distances show uniform (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	10 min = 600 s (1); 600 oscillations (1); T = 1 s (1).	1+1+1
16	8 min = 480 s (1); 3 x 10^5 x 480 (1); 1.44 x 10^8 km (approx.) (1).	1+1+1
17	To-and-fro (1); Repeats (1); Regular time (1).	1+1+1
	OR: 100 m in 90 s (1); 100/90 (1); 1.11 m/s (1).	1+1+1
18	Falling stone has gravity acting (1); Pendulum swings to and fro (1); Speed varies in both (1).	1+1+1
	OR: 30 min = 0.5 h (1); 36 x 0.5 = 18 km (1); 18000 m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $4\ \text{km}/\frac{1}{3}\ \text{h} = 12$ km/h (1). (ii) A horizontal line, as the distance does not change (1). (iii) $6\ \text{km}/\frac{1}{3}\ \text{h} = 18$ km/h (1). (iv) Total distance 10 km, total time 50 min $= \frac{5}{6}$ h (1); average speed $= 10 \div \frac{5}{6} = 12$ km/h (1).	1+1+1+2
20	Shadow-based sundials (1); Water clocks (ghati) (1); Sand and other timers (1); Astronomical observations (1); Calendars (1).	1+1+1+1+1
	OR: Passenger and platform (1); Speed relative to the ground (1); Relative to train is zero (1); Two moving cars (1); Frame matters (1).	1+1+1+1+1

Mistakes to avoid:

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- Writing that a heavier bob has a larger time period; the period depends on **length**, not on mass.
- Forgetting to convert units before calculating (minutes to hours, km/h to m/s).
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